

Alex Picard

Brookline, MA · (207) 551-5430 · am.picard03@gmail.com

<https://www.linkedin.com/in/alexpica0/> | <https://github.com/Alexm-picard> | <https://alexpica0.info>

EDUCATION

Boston University | MS in Software Development | 2025-2027

University of Maine | BS in Computer Engineering, Minor: Computer Science | 2021 - 2025

TECHNICAL SKILLS

Languages: Python, TypeScript, C/C++, Java, JavaScript

Tools/Cloud: Docker, AWS, React, Express, Flask, Node.js, PyTorch, TensorFlow, Git

Databases: NoSQL, Redis, SQLite, MongoDB

WORK EXPERIENCE

ML Data Associate II | Amazon

July 2025 - Present

- Engineer high-quality training datasets for LLMs by labeling and evaluating multimodal data across classification, ranking, and adversarial testing tasks, maintaining 94.5% quality score while processing 5,500+ samples and completing tasks 2.7x faster than standard
- Pilot test annotation workflows and UI tools pre-production, providing UI/UX feedback to improve interface design, annotator productivity, and global deployment success.

ML Research Assistant | Advanced Structures and Composite Center

May 2023 - September 2024

- Refactored over 3,000 lines of MATLAB code into modular Python components, enabling integration into an active learning pipeline for composite material simulation workflows
- Designed a multi-fidelity active learning system combining PINNs and feedforward neural networks in PyTorch to model composite materials, leveraging low-fidelity analytical models to reduce simulation computational cost.
- Led technical design and integration of neural network architecture within a 10-member interdisciplinary team, delivering quarterly milestones using Agile methodology

PROJECTS

Multi-Fidelity Physics Informed Neural Network (MF-PINN)

April 2025 - May 2025

- Developed dual-network PINN architecture in PyTorch, integrating low-fidelity analytical models and high-fidelity experimental data to predict concrete compressive strength, using physics-informed constraints to balance computational efficiency with accuracy.
- Implemented composite loss function combining physics residuals, low-fidelity data loss, and high-fidelity correction loss with LeakyReLU activation and dropout regularization, demonstrating proof-of-concept for multi-fidelity machine learning approaches.

Financial Planner Application | <https://financial-planner.alexpica0.info> **January 2025 - December 2025**

- Built a full-stack financial planning web application using React, TypeScript, Node.js/Express, and MongoDB with Auth0 authentication, featuring savings/debt tracking, automated payment scheduling, and transaction history for personal financial management.
- Collaborated in a 6-person team using GitHub Issues and pull requests to manage sprint deliverables, coordinate feature development, and conduct code reviews.

To-Do Webpage (Cloud) | <https://todo.alexpica0.info>

April 2025 - May 2025

- Architected and deployed a distributed microservices application using Docker Compose with 4 Flask-based services, 3 isolated Redis instances for data persistence, and Nginx reverse proxy for centralized routing and traffic management across services.
- Implemented RESTful APIs with inter-service communication using Docker's internal DNS resolution and deployed on AWS EC2 for production-grade hosting with industry-standard cloud deployment practices.